PDC

Personal Data Cleaner



Document privacy inside
your company’s infrastructure

Product overview and technical capabilities

a0nikolaevsoft@gmail.com

Personal and confidential data stays inside your company



Recognition and processing run on your servers. Documents are not sent to external cloud services.

Sensitive documents slow down everyday work

Legal review. QA data. Partner handoffs.

The operational problem

Teams must remove personal data before a document can leave its original context. Manual edits need repeated checks.

The product opportunity

A repeatable document workflow that keeps processing under the company’s control and records why replacements happened.

PDC turns files into a reusable working copy

Upload a file or a batch. Detect PII. Apply a replacement policy. Download the result.

PDC.Desktop + PDC.Web

Windows application and a browser interface for operational teams. Russian and English UI.

PDC.Server + API

Shared processing, administrator policies and integration endpoints. Server runs in the company environment.

Three ways to replace PII

Source document

Name: Pavel Sergeev
Phone: +7 912 405-18-73
Project: Northern Warehouse

1. Template replacement

Name: [TEMPLATE_FULLNAME]
Phone: [TEMPLATE_PHONENUMBER]
Project: Northern Warehouse

2. Data generation

Name: Alex Morgan
Phone: +1 202 555-0147
Project: Northern Warehouse

3. Combined mode

Name: [TEMPLATE_FULLNAME]
Phone: +1 202 555-0147
Project: Northern Warehouse

Illustrative data. Replacement modes can be combined by PII type.

First customers: teams sharing sensitive files

Segment	Initial workflow	Buyer
Legal and advisory	Prepare documents for external review	Practice lead / IT
Software teams	Prepare QA and support examples	Engineering / security
Finance and HR operations	Share working copies with contractors	Operations / security

Entry point: one workflow, a representative file set and an accountable business owner.

Positioning among existing approaches

Approach	What it offers	What the buyer still handles
Manual redaction	Direct reviewer control	Repeated edits and review
Custom scripts	Tailored business rules	Format handling and maintenance
Cloud services	Hosted processing	Data transfer and access policies
PDC	Files, local models, UI and API	Deployment and output validation

Qualitative workflow comparison.

A product for work inside the company

1.15.12.5

Current version on main

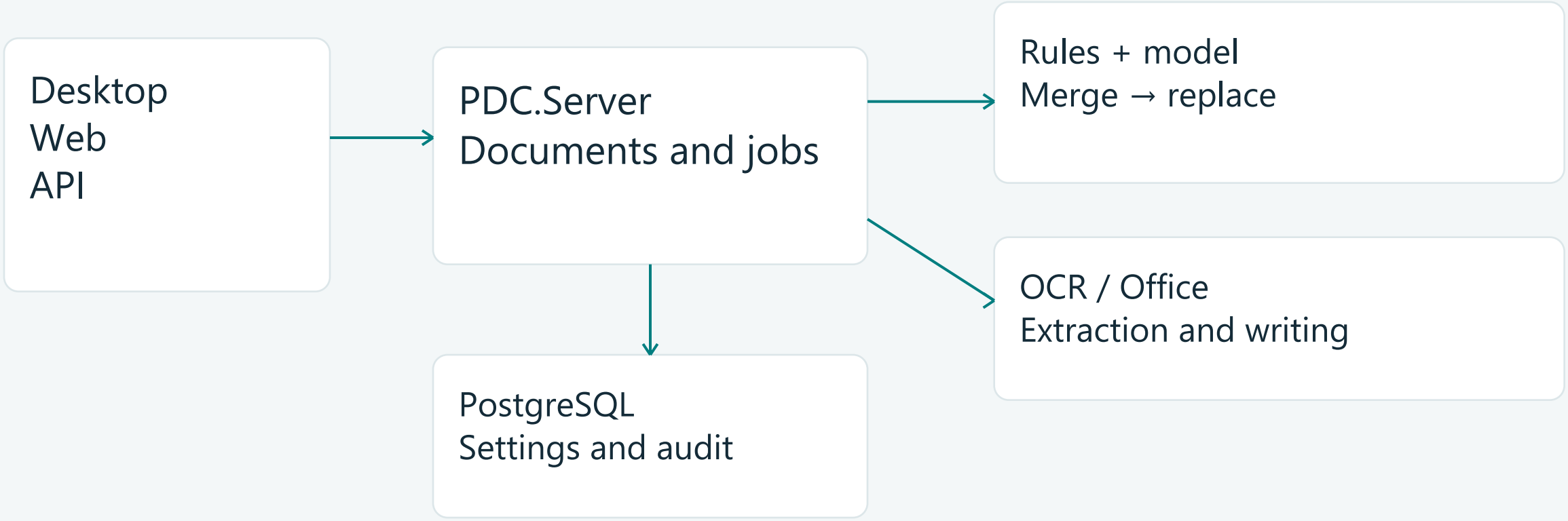
- Desktop — Windows application
- Web — browser interface
- API — files and background batches
- Shared policies and replacement reasons
- Processing on the company server

II

Technical specifications

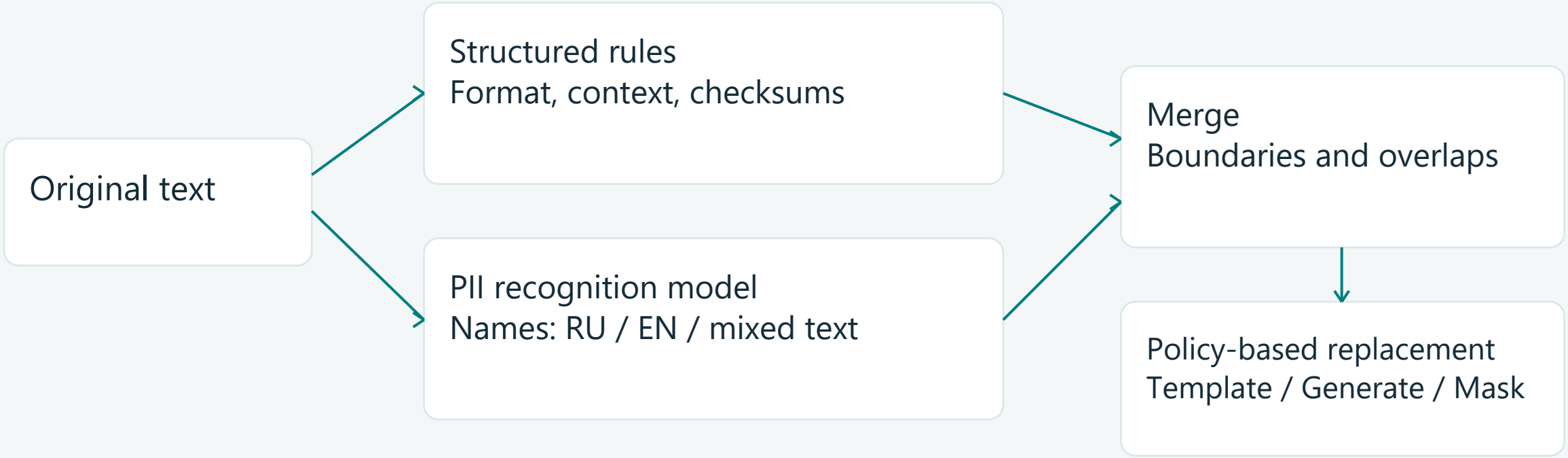
Architecture, formats, deployment and operating limits
Scoped measurements and remaining validation work.

Processing inside the company perimeter



Clients submit files and receive results. Data, processing services and storage stay inside the company perimeter.

Combined PII recognition



Both detectors use the original text before replacement. Full-name boundaries, initials and selection of the full name among overlapping fragments have improved.

Document format coverage

Family	Formats	Processing path
Office	DOCX, XLSX, PPTX	Native Open XML handlers
Legacy Office	DOC, XLS, PPT	Converter round trip
PDF	Text, scanned and mixed PDFs	Text extraction or page-level OCR
Text	TXT, CSV	Text replacement
Images	PNG, JPG/JPEG, BMP, TIF/TIFF	OCR and pixel replacement

Layout fidelity varies by format. PDF and legacy conversion have explicit limitations.

Integration uses files and background jobs

POST /api/documents/anonymize

One multipart file. Processed file in the HTTP response.

POST /api/documents/batch

202 + job ID. Poll status, download results, delete the job.
Database work uses a separate administrator-only jobs API.

Database processing has a defined safety scope

PostgreSQL / ClickHouse / SQL Server / Oracle

Implemented scope

Selected text columns in one table. Copy to a new table or explicit in-place processing. Administrator access and endpoint allowlist.

Operational constraints

Back up before in-place work. Control concurrent writes. Copies do not reproduce indexes, constraints or privileges. ClickHouse can report partial changes.

Security controls and the responsibility boundary

Implemented controls

JWT and User / Admin roles. Login rate limits. Same-origin Web sessions with HttpOnly refresh cookies. Local inference without a public model API.

Company responsibilities

HTTPS, secrets, storage access, backups and network policy. Review output for residual identifiers and metadata. Anonymization is not a compliance certificate.

Source bytes pass through server memory. Filenames, job metadata and replacement reasons can be stored. Processed-file caching is off by default.

PII recognition models

Russian

An experimental candidate is ready. New weights undergo validation before inclusion in the standard delivery.

English

Trials retained the original PII recognition model.

The historical compact candidate failed its quality gate; it does not replace the standard models.

Adding a new language requires model fine-tuning and updated regular expressions.

Compact: resource savings in the fixed protocol

Observation	default / 007	compact / 008
Sampled HTTP working-set peak	8.868 GiB	1.097 GiB
Service CPU / HTTP success	6,903.690 s; 516/530	454.929 s; 530/530
Warm RU TXT p50 / p95	2.481 / 2.912 s	0.031 / 0.051 s
Warm EN TXT p50 / p95	4.271 / 10.673 s	0.017 / 0.036 s

24 repeats/file at C1. Same fixtures; failures retained. CPU has unequal success mixes. Sampled RAM excludes host/cache; compact physical minimum untested.

Historical budgets and deployment planning

Parameter	minimal / 005	standard / 004
Test CPU / RAM quota	2 CPU / 10 GiB, swap 0	4 CPU / 12 GiB, swap 0
Sampled stack peak	8.917 GiB working set	8.718 GiB working set
VM planning allowance	2 vCPU / 16 GiB RAM	4 vCPU / 24 GiB RAM
Free SSD allowance	40 GiB, planning only	60 GiB, planning only

Test quotas exclude host/VM overhead. VM/SSD allowances are engineering guidance. Minimal did not reduce the measured active RAM peak.

Historical latency and reliability

Case / observation	standard	minimal
RU TXT p50 / p95	2.037 / 2.428 s	3.184 / 4.024 s
EN TXT p50 / p95	2.157 / 17.893 s	3.429 / 29.935 s
Controlled HTTP success	1,920 / 1,920	1,920 / 1,920
External PDF / stress C8+	38/50; 102/144 success	34/50; 40/144 success

Repeated EN C1: 6.62–9.52 vs 4.27–6.67 docs/min. Same fixtures, result cache off. Start with C1; C8/16/48 stress is outside the operating envelope.

B3 / repeated C1 cohorts: 96 requests per file; v1.15.5.1

Historical startup, idle memory and storage

Observation	standard	minimal
Ready after restart	23.844–31.969 s	23.672–32.531 s
First document	0.976–28.495 s; 12/12	1.132–50.328 s; 12/12
EN after 6 min idle	Remains loaded (006)	Unloaded; reload 25.969 s
Image inspect / model volumes	5.661 GiB packed content	RU 2.648 / EN 2.027 GiB

Model files persisted; OS cache uncontrolled. Minimal PII recognition model: 4.415 to 0.721 GiB after idle. Inspect omits unpacked data on this engine; it is not installed disk usage.

Detection and document limits

Checks and measurements

Current source: 1,030 backend tests. Measured v1.15.12.0: 3,308 texts, zero call errors. External enabled F1: EN 76.24%, RU 89.54%. External sources are synthetic.

Observed limits

22-file API: 20 outputs and 2 explicit refusals. Office authors and previews corrected. Name misses remain. The new four-text sample contains one missed name.

Windows installer Authenticode signing and arbitrary local model directories remain separate open tasks.

Combined recognition quality

4,256 synthetic development texts · model + structured rules

Language	Precision	Recall	F1
RU	98.96%	100%	99.48%
EN	100%	100%	100%
RU + EN	98.96%	100%	99.48%

27,792 matched occurrences · 0 misses · 240 false positives

Calculated from exact saved model responses on the own development set. This is not the share of fully cleaned documents or a guarantee for new files, Office, PDF or OCR.